

Foundation (Jalapeno)

Q1. Factor the expression: $x^2 + 3x + 2x + 6$

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 6)$
- (C) $(x + 3)(x + 2)$
- (D) $(x + 2)(x + 1)$
- (E) $(x + 6)(x + 1)$

Q2. Factor: $2x^2 + 5x + 2x + 5$

- (A) $(2x + 1)(x + 5)$
- (B) $(x + 5)(2x + 1)$
- (C) $(2x + 5)(x + 1)$
- (D) $(x + 1)(2x + 5)$
- (E) $(2x + 2)(x + 2)$

Q3. A company has x dollars invested in stocks and $2x$ dollars in bonds. If they earn x dollars from stocks and $3x$ dollars from bonds, factor the total earnings: $x + 2x + x + 3x$

- (A) $7x$
- (B) $6x$
- (C) $5x$
- (D) $4x$
- (E) $3x$

Q4. Factor the expression: $x^2 + 2x + 5x + 10$

- (A) $(x + 2)(x + 5)$
- (B) $(x + 5)(x + 2)$
- (C) $(x + 1)(x + 10)$
- (D) $(x + 10)(x + 1)$
- (E) $(x + 5)(x + 1)$

Q5. Factor: $3x^2 + 2x + 6x + 4$

- (A) $(3x + 2)(x + 2)$
- (B) $(x + 2)(3x + 2)$
- (C) $(3x + 6)(x + 1)$
- (D) $(x + 1)(3x + 6)$
- (E) $(3x + 6)(x + 2)$

Q6. A savings account earns $2x$ dollars in interest and has $3x$ dollars deposited into it. Factor the total amount: $2x + 3x + x + 2x$

- (A) $8x$
- (B) $6x$
- (C) $5x$
- (D) $4x$
- (E) $3x$

Q7. Factor the expression: $x^2 + x + 4x + 4$

- (A) $(x + 1)(x + 4)$
- (B) $(x + 4)(x + 1)$
- (C) $(x + 2)(x + 2)$
- (D) $(x + 2)(x + 3)$
- (E) $(x + 3)(x + 2)$

Q8. Factor: $x^2 + 2x + 2x + 4$

- (A) $(x + 2)(x + 2)$
- (B) $(x + 2)(x + 4)$
- (C) $(x + 1)(x + 4)$
- (D) $(x + 4)(x + 1)$
- (E) $(x + 4)(x + 2)$

Open Ended Questions (FRQ)

Q9. Factor the expression: $x^2 + 5x + 3x + 15$. Verify your answer by multiplying.

Q10. A stock portfolio has x dollars invested in tech stocks, $2x$ dollars in healthcare stocks, x dollars in finance stocks, and $3x$ dollars in energy stocks. Factor the total value of the portfolio: $x + 2x + x + 3x$. Verify your answer.

Chef's Tips

- Provide clear instructions for each question type
- Ensure students have necessary tools (e.g. calculators, pencils)
- Allow students to ask questions before starting the test